

Lec-05-Week-03

Wednesday, 19 August 2026

4:47 pm

$$(x, y)$$

$$(r \cos \theta, r \sin \theta)$$



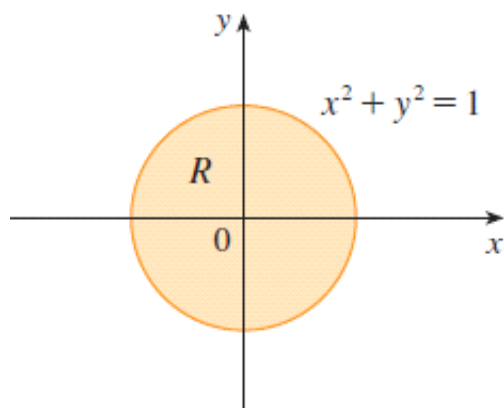
POLAR BEAR



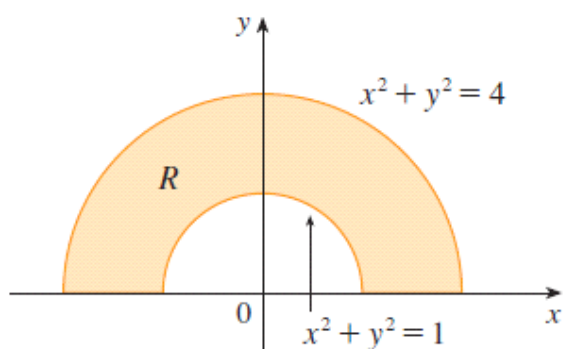
CARTESIAN BEAR



Double Integrals in Polar Coordinates



$$R = \{(r, \theta) \mid 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$$



$$(b) R = \{(r, \theta) \mid 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}$$

$$r^2 = x^2 + y^2 \quad x = r \cos \theta \quad y = r \sin \theta$$

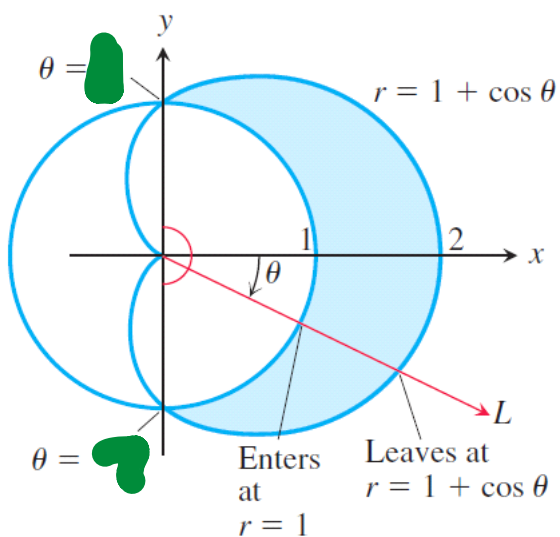
The regions in Figure 1 are special cases of a **polar rectangle**

$$R = \{(r, \theta) \mid a \leq r \leq b, \alpha \leq \theta \leq \beta\}$$

2 Change to Polar Coordinates in a Double Integral If f is continuous on a polar rectangle R given by $0 \leq a \leq r \leq b$, $\alpha \leq \theta \leq \beta$, where $0 \leq \beta - \alpha \leq 2\pi$, then

$$\iint_R f(x, y) \, dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta$$

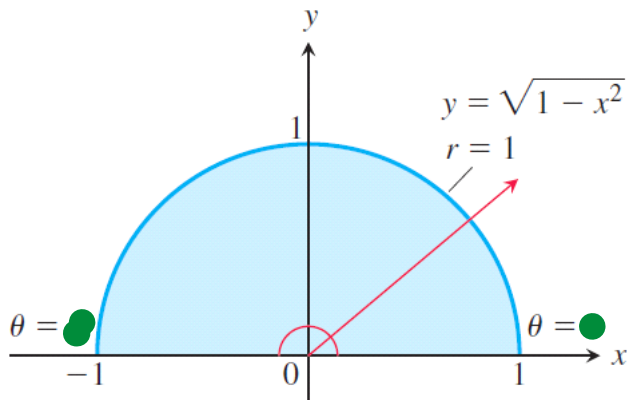
EXAMPLE 1 Find the limits of integration for integrating $f(r, \theta)$ over the region R that lies inside the cardioid $r = 1 + \cos \theta$ and outside the circle $r = 1$.



EXAMPLE 3 Evaluate

$$\iint_R e^{x^2+y^2} dy dx,$$

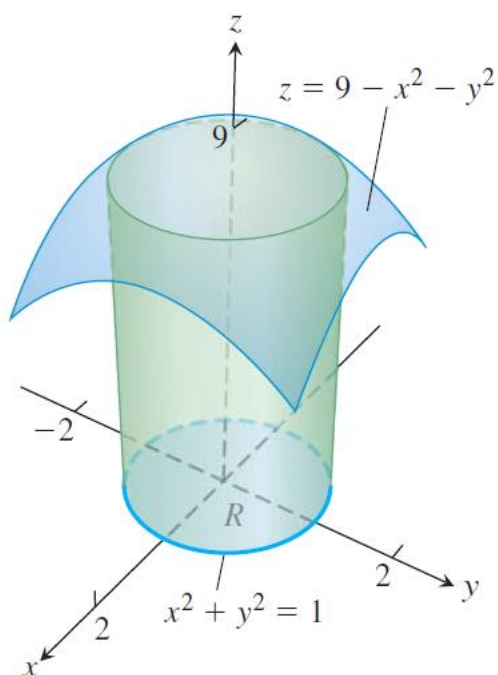
where R is the semicircular region bounded by the x -axis and the curve $y = \sqrt{1 - x^2}$ (Figure 15.27).



$$\begin{aligned} \iint_R e^{x^2+y^2} dy dx &= \int_0^\pi \int_0^1 e^{r^2} r dr d\theta = \int_0^\pi \left[\frac{1}{2} e^{r^2} \right]_0^1 d\theta \\ &= \int_0^\pi \frac{1}{2} (e - 1) d\theta = \frac{\pi}{2} (e - 1). \end{aligned}$$

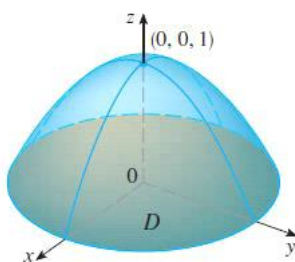
EXAMPLE 4 Evaluate the integral

$$\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx.$$



$$\begin{aligned} \int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx &= \int \int (r^2) r dr d\theta \\ &= \int_0^{\pi/2} \left[\frac{r^4}{4} \right]_{r=0}^{r=1} d\theta = \int_0^{\pi/2} \frac{1}{4} d\theta = \frac{\pi}{8}. \end{aligned}$$

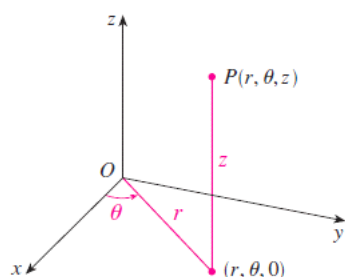
V EXAMPLE 2 Find the volume of the solid bounded by the plane $z = 0$ and the paraboloid $z = 1 - x^2 - y^2$.



SOLUTION If we put $z = 0$ in the equation of the paraboloid, we get $x^2 + y^2 = 1$. This means that the plane intersects the paraboloid in the circle $x^2 + y^2 = 1$, so the solid lies under the paraboloid and above the circular disk D given by $x^2 + y^2 \leq 1$ [see Figures 6 and 1(a)]. In polar coordinates D is given by $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$. Since $1 - x^2 - y^2 = 1 - r^2$, the volume is

$$\begin{aligned} V &= \iint_D (1 - x^2 - y^2) dA = \int_0^{2\pi} \int_0^1 (1 - r^2) r dr d\theta \\ &= \int_0^{2\pi} d\theta \int_0^1 (r - r^3) dr = 2\pi \left[\frac{r^2}{2} - \frac{r^4}{4} \right]_0^1 = \frac{\pi}{2} \end{aligned}$$

Triple Integrals in Cylindrical Coordinates



$$x = r \cos \theta \quad y = r \sin \theta \quad z = z$$

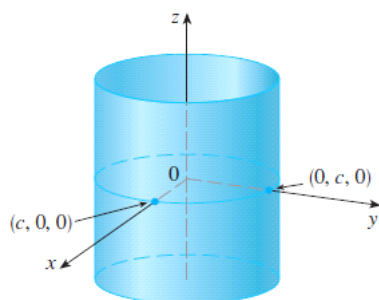


FIGURE 4
 $r = c$, a cylinder

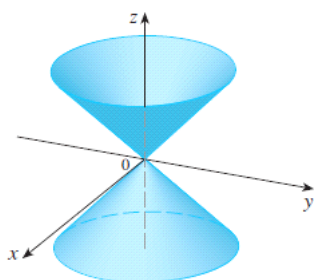


FIGURE 5
 $z = r$, a cone

Equations Relating Rectangular (x, y, z) and Cylindrical (r, θ, z) Coordinates

$$\begin{aligned} x &= r \cos \theta, & y &= r \sin \theta, & z &= z, \\ r^2 &= x^2 + y^2, & \tan \theta &= y/x \end{aligned}$$

$r = 4$ Cylinder, radius 4, axis the z -axis

$\theta = \frac{\pi}{3}$ Plane containing the z -axis

$z = 2$. Plane perpendicular to the z -axis

$$\iiint_E f(x, y, z) dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos \theta, r \sin \theta)}^{u_2(r \cos \theta, r \sin \theta)} f(r \cos \theta, r \sin \theta, z) r dz dr d\theta$$

[Cylindrical Coordinates: Dynamic Illustrator](#)

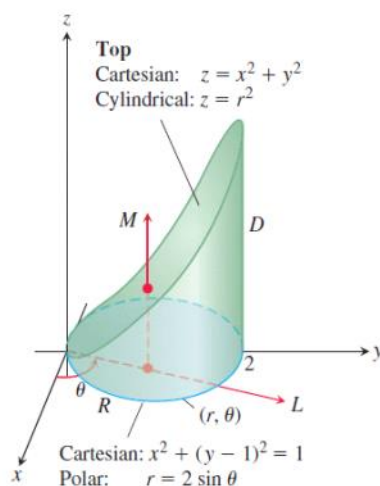
Evaluate the triple integral

$$\iiint_B (zr \sin \theta) r dr d\theta dz$$

where the cylindrical box B is $B = \{(r, \theta, z) | 0 \leq r \leq 2, 0 \leq \theta \leq \pi/2, 0 \leq z \leq 4\}$.

$$\begin{aligned} \int_{\theta=0}^{\theta=\pi/2} \int_{r=0}^{r=2} \int_{z=0}^{z=4} (zr \sin \theta) r dz dr d\theta &= \left(\int_0^{\pi/2} \sin \theta d\theta \right) \left(\int_0^2 r^2 dr \right) \left(\int_0^4 z dz \right) \\ &= \left(-\cos \theta \Big|_0^{\pi/2} \right) \left(\frac{r^3}{3} \Big|_0^2 \right) \left(\frac{z^2}{2} \Big|_0^4 \right) = \frac{64}{3}. \end{aligned}$$

EXAMPLE 1 Find the limits of integration in cylindrical coordinates for integrating a function $f(r, \theta, z)$ over the region D bounded below by the plane $z = 0$, laterally by the circular cylinder $x^2 + (y - 1)^2 = 1$, and above by the paraboloid $z = x^2 + y^2$.

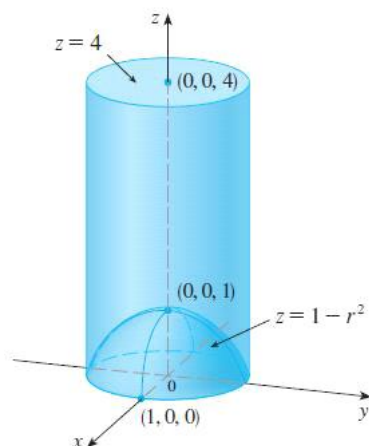


$$\begin{aligned} x^2 + (y - 1)^2 &= 1 \\ x^2 + y^2 - 2y + 1 &= 1 \\ r^2 - 2r \sin \theta &= 0 \\ r &= 2 \sin \theta. \end{aligned}$$

$$\iiint_D f(r, \theta, z) dV = \int_0^{\pi} \int_0^{2 \sin \theta} \int_0^{r^2} f(r, \theta, z) dz r dr d\theta.$$

V EXAMPLE 3 A solid E lies within the cylinder $x^2 + y^2 = 1$, below the plane $z = 4$, and above the paraboloid $z = 1 - x^2 - y^2$. (See Figure 8.) The density at any point is proportional to its distance from the axis of the cylinder. Find the mass of E .

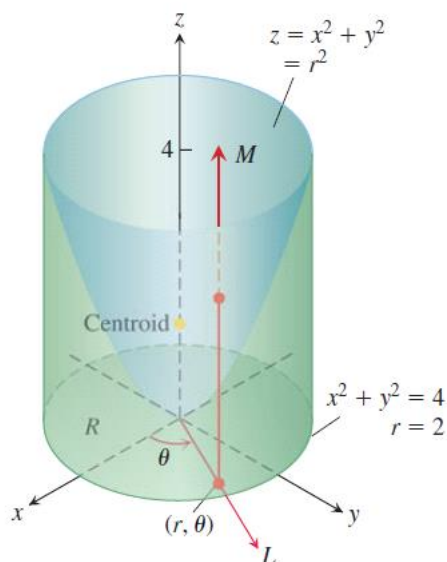
$$f(x, y, z) = K\sqrt{x^2 + y^2} = Kr$$



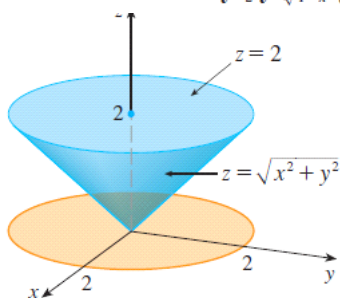
$$E = \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, 0 \leq r \leq 1, 1 - r^2 \leq z \leq 4\}$$

$$\begin{aligned} m &= \iiint_E K\sqrt{x^2 + y^2} \, dV = \int_0^{2\pi} \int_0^1 \int_{1-r^2}^4 (Kr) \, r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^1 Kr^2 [4 - (1 - r^2)] \, dr \, d\theta = K \int_0^{2\pi} d\theta \int_0^1 (3r^2 + r^4) \, dr \\ &= 2\pi K \left[r^3 + \frac{r^5}{5} \right]_0^1 = \frac{12\pi K}{5} \end{aligned}$$

EXAMPLE 2 Find the centroid ($\delta = 1$) of the solid enclosed by the cylinder $x^2 + y^2 = 4$, bounded above by the paraboloid $z = x^2 + y^2$, and bounded below by the xy -plane.



EXAMPLE 4 Evaluate $\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2 + y^2) \, dz \, dy \, dx$.

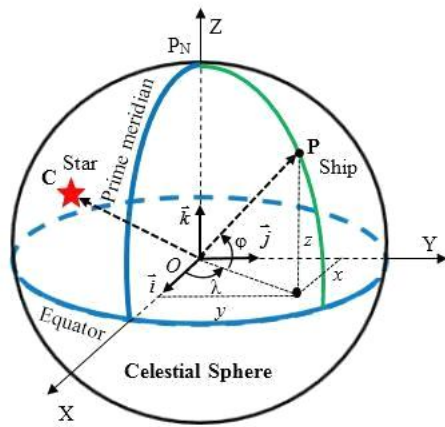


$$E = \{(x, y, z) \mid -2 \leq x \leq 2, -\sqrt{4-x^2} \leq y \leq \sqrt{4-x^2}, \sqrt{x^2+y^2} \leq z \leq 2\}$$

$$E = \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, 0 \leq r \leq 2, r \leq z \leq 2\}$$

$$\begin{aligned} \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2 + y^2) \, dz \, dy \, dx &= \iiint_E (x^2 + y^2) \, dV \\ &= \int_0^{2\pi} \int_0^2 \int_r^2 r^2 \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} d\theta \int_0^2 r^3(2-r) \, dr \\ &= 2\pi \left[\frac{1}{2}r^4 - \frac{1}{5}r^5 \right]_0^2 = \frac{16}{5}\pi \end{aligned}$$

Triple Integrals in Spherical Coordinates



The importance of using the proper coordinate system

$$f(\rho, \theta, \phi)$$



$$f(x, y, z)$$

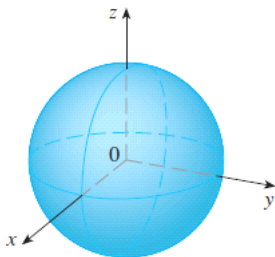


FIGURE 2 $\rho = c$, a sphere

$$\rho^2 = x^2 + y^2 + z^2$$

$$E = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}$$

DEFINITION Spherical coordinates represent a point P in space by ordered triples (ρ, ϕ, θ) in which

1. ρ is the distance from P to the origin ($\rho \geq 0$).
2. ϕ is the angle $\vec{OP}$ makes with the positive z -axis ($0 \leq \phi \leq \pi$).
3. θ is the angle from cylindrical coordinates.

Equations Relating Spherical Coordinates to Cartesian and Cylindrical Coordinates

$$\begin{aligned} r &= \rho \sin \phi, & x &= r \cos \theta = \rho \sin \phi \cos \theta, \\ z &= \rho \cos \phi, & y &= r \sin \theta = \rho \sin \phi \sin \theta, \\ \rho &= \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2}. \end{aligned}$$

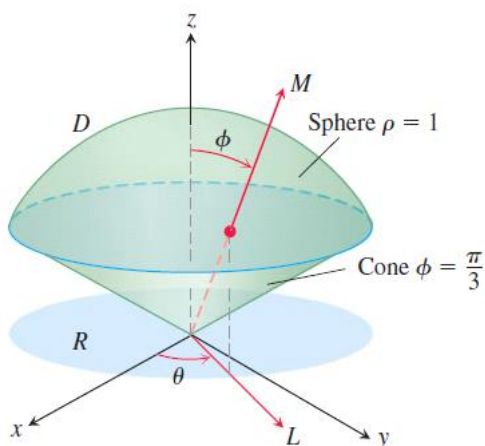
$$\begin{aligned} \text{[3]} \quad \iiint_E f(x, y, z) \, dV &= \int_c^d \int_\alpha^\beta \int_a^b f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi \end{aligned}$$

where E is a spherical wedge given by

$$E = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}$$

[Spherical Coordinates: Dynamic Illustrator](#)

EXAMPLE 5 Find the volume of the “ice cream cone” D cut from the solid sphere $\rho \leq 1$ by the cone $\phi = \pi/3$.



$$\begin{aligned} V &= \iiint_D \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/3} \int_0^1 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= \int_0^{2\pi} \int_0^{\pi/3} \left[\frac{\rho^3}{3} \right]_0^1 \sin \phi \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/3} \frac{1}{3} \sin \phi \, d\phi \, d\theta \\ &= \int_0^{2\pi} \left[-\frac{1}{3} \cos \phi \right]_0^{\pi/3} d\theta = \int_0^{2\pi} \left(-\frac{1}{6} + \frac{1}{3} \right) d\theta = \frac{1}{6} (2\pi) = \frac{\pi}{3}. \end{aligned}$$

V EXAMPLE 3 Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{3/2}} dV$, where B is the unit ball:

$$B = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$$

$$B = \{(\rho, \theta, \phi) \mid 0 \leq \rho \leq 1, 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \pi\}$$

In addition, spherical coordinates are appropriate because

$$x^2 + y^2 + z^2 = \rho^2$$

Thus [3] gives

$$\begin{aligned} \iiint_B e^{(x^2+y^2+z^2)^{3/2}} dV &= \int_0^\pi \int_0^{2\pi} \int_0^1 e^{(\rho^2)^{3/2}} \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi \\ &= \int_0^\pi \sin \phi \, d\phi \int_0^{2\pi} d\theta \int_0^1 \rho^2 e^{\rho^3} \, d\rho \\ &= [-\cos \phi]_0^\pi (2\pi) \left[\frac{1}{3}e^{\rho^3}\right]_0^1 = \frac{4}{3}\pi(e - 1) \end{aligned}$$

EXAMPLE 6 A solid of constant density $\delta = 1$ occupies the region D in Example 5. Find the solid's moment of inertia about the z -axis.

Solution In rectangular coordinates, the moment is

$$I_z = \iiint_D (x^2 + y^2) \, dV.$$

Coordinate Conversion Formulas

CYLINDRICAL TO RECTANGULAR

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \\ z &= z \end{aligned}$$

SPHERICAL TO RECTANGULAR

$$\begin{aligned} x &= \rho \sin \phi \cos \theta \\ y &= \rho \sin \phi \sin \theta \\ z &= \rho \cos \phi \end{aligned}$$

SPHERICAL TO CYLINDRICAL

$$\begin{aligned} r &= \rho \sin \phi \\ z &= \rho \cos \phi \\ \theta &= \theta \end{aligned}$$

Corresponding formulas for dV in triple integrals:

$$\begin{aligned} dV &= dx \, dy \, dz \\ &= dz \, r \, dr \, d\theta \\ &= \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \end{aligned}$$

